

CROSS-CUTTING · EXAM STRATEGY - ALL DOMAINS

Monitoring, Logging and Observability

Knowing what your architecture is actually doing.

COVERS ALL DOMAINS SCENARIOS 8 MODULE 13 of 22

CloudWatch sees what AWS can see from **outside** the instance. Anything inside it needs the agent.

The CloudWatch components

COMPONENT	WHAT IT DOES	EXAM SIGNAL
Metrics	Numeric time series. 5-minute default, 1-minute detailed	'Track utilisation over time'
Alarms	Watch a metric, act when it breaches	'Notify when', 'automatically restart'
Composite alarms	Combine several alarms with AND or OR logic	'Only alert when both conditions'
Logs	Log storage with configurable retention	'Centralise application logs'
Metric filters	Turn a log pattern into a metric you can alarm on	'Alert when this appears in logs'
Logs Insights	Ad-hoc query language over log data	'Investigate', 'search across logs'
Synthetics canaries	Scripted checks against an endpoint from outside	'Detect an outage before users do'
Contributor Insights	Finds the top talkers in log data	'Which client is causing this'

ON THE EXAM

A metric filter creates an alarm from logs. Logs Insights is for humans investigating. They are not interchangeable.

Metric filter or Logs Insights?

METRIC FILTER

Automated and continuous

- Runs on every incoming log event
- Produces a metric, so you can alarm on it
- Set up once, fires forever
- For conditions you know in advance

LOGS INSIGHTS

Interactive and investigative

- Run by a person, when they ask a question
- Returns query results, not metrics
- Cannot trigger an alarm
- For questions you did not anticipate

THE TIEBREAKER

'Automatically notify when X appears in the logs' is a metric filter. 'Work out why last Tuesday went wrong' is Logs Insights.

Beyond CloudWatch

- ◆ **AWS X-Ray.** Distributed tracing across services. Builds a service map and shows where latency accumulates.
- ◆ **CloudWatch RUM.** Real user monitoring from actual browsers, for client-side performance.
- ◆ **AWS Health Dashboard.** Tells you about AWS-side events affecting your specific resources.
- ◆ **EventBridge Scheduler.** The modern answer for scheduled invocation. Replaces cron on an instance.
- ◆ **Systems Manager.** Patch Manager, Run Command and inventory across a fleet without SSH.

ON THE EXAM

'Which microservice in the chain is slow' is X-Ray. CloudWatch tells you a service is slow; X-Ray tells you which one.

An operations team wants to be alerted when memory utilisation on a fleet of EC2 instances exceeds eighty percent. They have opened the CloudWatch console and cannot find a memory metric.

They ask whether memory monitoring requires a third-party tool.

What should the solutions architect tell them?

- ☐ A Enable detailed monitoring on the instances to expose memory metrics
- ☐ B Install the CloudWatch agent to publish memory as a custom metric
- ☐ C Memory cannot be monitored on EC2 without a third-party tool
- ☐ D Use VPC Flow Logs to infer memory pressure

◆ PAUSE AND DECIDE

What should the solutions architect tell them?

- A** Detailed monitoring

Increases metric frequency from five minutes to one. It does not add new metric types, and memory is still absent.
- C** Needs a third-party tool

An invented restriction. The CloudWatch agent does exactly this.

- B** CloudWatch agent

The agent runs inside the operating system, where memory is visible, and publishes it as a custom metric you can alarm on.
- D** Flow Logs

Record network traffic metadata. They say nothing about memory.

THE SIGNAL

Memory and disk space are inside the OS, so they need the agent. Detailed monitoring changes frequency, never metric types.

A microservices application makes calls across nine services. Users report that some checkout requests take eight seconds while most complete in under one.

CloudWatch shows every individual service responding within its normal range, and the team cannot determine where the time is going.

Which service should be used to identify the cause?

- ☐ A AWS X-Ray with tracing enabled across the services
- ☐ B CloudWatch Logs Insights across all nine log groups
- ☐ C CloudWatch detailed monitoring on every service
- ☐ D VPC Flow Logs between the services

◆ PAUSE AND DECIDE

Which service should be used to identify the cause?

- A

 X-Ray tracing
Follows one request end to end across all nine services, showing exactly where the eight seconds accumulated.
- B

 Logs Insights
Can query each service's logs but cannot correlate one slow request across nine of them without a shared trace.
- C

 Detailed monitoring
More frequent averages. Averages are precisely what is hiding a slow minority of requests.
- D

 Flow Logs
Record which traffic flowed, not how long application processing took.

THE SIGNAL

Every service looks healthy individually but the whole is slow. That is the distributed tracing signature.

A security team must be notified automatically whenever the phrase 'authentication failure' appears more than fifty times in five minutes in an application's log group.

They do not want a person running queries to discover this.

Which approach meets this requirement?

- ☐ A A scheduled CloudWatch Logs Insights query run every five minutes
- ☐ B A metric filter on the log group with a CloudWatch alarm and SNS
- ☐ C Amazon GuardDuty monitoring the log group
- ☐ D A CloudTrail trail filtered on authentication events

◆ PAUSE AND DECIDE

Which approach meets this requirement?

- A

Scheduled Insights query

Insights returns results to a person or a script. Building scheduling and alerting around it recreates what a metric filter already does.
- B

Metric filter, alarm, SNS

The filter turns the log pattern into a metric, the alarm applies the threshold and window, and SNS delivers the notification.
- C

GuardDuty

Analyses AWS-generated logs such as CloudTrail and DNS logs for threats. It does not watch your application's custom log group.
- D

CloudTrail filter

CloudTrail records AWS API calls. Application-level authentication failures are not AWS API calls.

THE SIGNAL

Turning a log pattern into an alarm is the definition of a metric filter. Insights cannot trigger anything.

An on-call team is being woken by alerts every time a single instance in a large Auto Scaling group breaches its CPU alarm, even though the service remains healthy and the group replaces the instance automatically.

They want to be paged only when high CPU coincides with the load balancer reporting unhealthy targets.

What should be configured?

- ☐ A A composite alarm combining the CPU alarm and the unhealthy host alarm with AND
- ☐ B Raise the CPU alarm threshold until it stops firing
- ☐ C Increase the alarm's evaluation periods to twelve
- ☐ D Move the notifications to a separate SNS topic nobody subscribes to

◆ PAUSE AND DECIDE

What should be configured?

- A

 Composite alarm with AND
Fires only when both conditions hold, which is exactly the stated requirement and the reason composite alarms exist.
- B

 Raise the threshold
Reduces noise by also missing genuine incidents. It trades one failure for a worse one.
- C

 More evaluation periods
Delays every alert including real ones, and a sustained single-instance breach would still page the team.
- D

 Unsubscribed topic
Silences the alert entirely rather than making it meaningful.

THE SIGNAL

'Only alert when two things are both true' is the composite alarm signature.

A company's checkout page was unavailable for forty minutes overnight. Instances were healthy, CloudWatch metrics looked normal, and nobody noticed until customers complained the next morning. The failure was a misconfigured third-party payment script that broke page loading.

The team wants to detect this class of failure automatically in future.

Which solution should be used?

- ☐ A A CloudWatch Synthetics canary running the checkout journey on a schedule
- ☐ B A CloudWatch alarm on the load balancer's request count
- ☐ C Enable detailed monitoring on the web instances
- ☐ D A metric filter on the web server access logs

◆ PAUSE AND DECIDE

Which solution should be used?

- A

 Synthetics canary
Scripts the actual user journey from outside and fails when the page does not work, regardless of how healthy the infrastructure looks.
- B

 Request count alarm
Requests were still arriving and being served. The infrastructure metric looked normal throughout, which is the whole problem.
- C

 Detailed monitoring
More frequent infrastructure metrics, which were never the failing signal.
- D

 Metric filter on access logs
The server returned pages successfully. The breakage was client-side, so the access logs look healthy.

THE SIGNAL

Infrastructure healthy but the user journey broken is the canary signature. Every other option watches the infrastructure.

An operations team runs a nightly report by leaving a cron job on a small EC2 instance that exists solely to trigger a Lambda function at 02:00.

The instance was recently terminated by accident and the reports silently stopped for eleven days. The team wants to remove this dependency.

What should replace the cron job?

- ☐ A An Amazon EventBridge scheduled rule invoking the Lambda function
- ☐ B A second EC2 instance running the same cron job for redundancy
- ☐ C An Auto Scaling group with a minimum of one to keep the instance alive
- ☐ D A CloudWatch alarm that triggers the function when metrics are low

◆ PAUSE AND DECIDE

What should replace the cron job?

- A

 EventBridge scheduled rule
Managed scheduling with no instance at all, so there is nothing to terminate and nothing to patch.
- B

 Second instance
Two instances to maintain, and the job would then run twice every night.
- C

 Auto Scaling minimum of one
Keeps an instance running, but you still own and patch a server whose only purpose is to run a clock.
- D

 Metric-triggered alarm
Alarms respond to metric conditions, not to a time of day.

THE SIGNAL

An instance whose only job is to run a schedule is always the wrong answer. EventBridge does it with nothing running.

A company must patch two thousand EC2 instances across twelve accounts on a monthly schedule, produce evidence of compliance, and do so without opening inbound ports or distributing SSH keys.

Instances run a mixture of Linux and Windows.

Which solution meets these requirements?

- ☐ A AWS Systems Manager Patch Manager with maintenance windows
- ☐ B A bastion host with an Ansible playbook run monthly
- ☐ C Rebuild all instances monthly from a new AMI
- ☐ D AWS Config rules reporting on patch levels

◆ PAUSE AND DECIDE

Which solution meets these requirements?

- A** Patch Manager with maintenance windows
Agent-based so no inbound ports or keys, works across Linux and Windows and many accounts, schedules via maintenance windows and reports compliance.
- B** Bastion and playbook
Requires exactly the open ports and distributed keys the question forbids.
- C** Monthly rebuild
A valid immutable strategy in principle, and an enormous undertaking for two thousand mixed instances with no compliance reporting.
- D** Config rules
Reports whether patching happened. It does not perform the patching.

THE SIGNAL

No inbound ports, no keys, plus fleet-wide action means Systems Manager. Config reports; it never acts.

A team wants to store application logs from three hundred instances centrally, retain them for thirteen months for audit, and keep the cost as low as possible while still allowing ad-hoc investigation.

Logs are currently written to local disk on each instance and lost when instances are replaced.

Which approach meets these requirements?

- ☐ A CloudWatch agent to CloudWatch Logs with thirteen-month retention, exporting older data to S3
- ☐ B Keep logs on local disk and increase the EBS volume size
- ☐ C Write logs to an EFS file system mounted on all instances
- ☐ D Send logs to CloudWatch Logs with indefinite retention

◆ PAUSE AND DECIDE

Which approach meets these requirements?

- A

Agent to Logs, retention set, export to S3

Centralises logs, survives instance replacement, allows Insights queries, and moves older data to cheaper storage.
- B

Bigger local volumes

Logs are still lost when an instance is replaced, and nothing is centralised or queryable.
- C

EFS for logs

Centralises the files but provides no query capability, no retention management, and three hundred instances writing to one file system is a poor pattern.
- D

Indefinite retention

Meets the audit requirement while ignoring cost. CloudWatch Logs storage is more expensive than S3 for long-term retention.

THE SIGNAL

The default retention in CloudWatch Logs is 'never expire'. Setting it is one of the easiest cost savings available.

Module 13 takeaways

- ◆ **Memory and disk space need the CloudWatch agent.** Detailed monitoring changes frequency, not metric types.
- ◆ **Metric filter alarms automatically.** Logs Insights is a human investigating.
- ◆ **Every service healthy but the whole request slow** means X-Ray.
- ◆ **Infrastructure healthy but the user journey broken** means a Synthetics canary.
- ◆ **An instance whose only job is a schedule** should be EventBridge.

ON THE EXAM

Next: migration and hybrid architecture, where the answer usually depends on bandwidth, downtime tolerance, or what cannot be changed.